

Ovarian cancer detection and classification using attention-based CNN and deep feature extraction techniques

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Abstract

Ovarian cancer (OC) remains a significant global health concern, marked by high mortality rates and a lack of reliable diagnostic tools. Early detection of this disorder is crucial for improving survival rates and optimizing the healthcare system. However, it is necessary to establish an automated system that is more accurate and consistent for decision-making. In this paper, we proposed an attention-based CNN model for classifying ovarian cancer. The proposed model is evaluated on the STRAMPN dataset, which comprises 987 histopathological images. Initially, various preprocessing techniques were extensively analyzed to determine the most suitable image representation, and data augmentation schemes were applied to enhance model robustness. After that, a modified attention-based ResNet50 architecture was proposed and fine-tuned through transfer learning to extract meaningful features from the training samples. Then the SHAP method was employed for feature selection to determine the most pertinent features. Finally, we utilized an attention-based CNN model to classify the images of OC. The proposed model achieved an accuracy of 99.09%, precision of 99.17%, recall of 99.21%, F1-score of 99.06%, and an AUC of 99.85%, with a training time of 1.93 seconds, demonstrating strong computational efficiency. Experimental results demonstrate that the proposed model outperforms existing methods based on quantitative evaluation. Thus, it could assist clinicians in diagnosing OC in clinical contexts. Furthermore, the proposed model is incorporated into a web application utilizing the FastAPI framework to facilitate real-time predictions. The web application can be accessed through the following link: <https://ovarian.franciscrudra.com>.

Keywords Ovarian cancer · Transfer learning · Feature extraction · Deep learning · Attention mechanism · Convolutional neural networks · Explainable AI

Abbreviations

OC	Ovarian cancer
CNN	Convolutional neural network
AUC	Area under the curve
AI	Artificial intelligence
ML	Machine learning

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DL	Deep learning
CT	Computed tomography
MRI	Magnetic resonance imaging
ANN	Artificial neural network
ReLU	Rectified linear unit
MCC	Matthew's correlation coefficient
PCA	Principal component analysis
ROC	Receiver-operating characteristic curve
SHAP	SHapley additive explanations

Introduction

Ovarian cancer ranks among the three leading gynecological cancers that critically endanger women's health worldwide¹. Although the fatality rate of ovarian cancer has declined in recent years, it still holds the second-highest position among gynecological malignancies, following uterine cancer². It arises from alterations in genetic material (DNA), leading to the abnormal and accelerated growth of ovarian cells, which can infiltrate and damage surrounding healthy tissues³. Additionally, ovarian cancer can result from inherited genetic mutations, particularly in the BRCA1 and BRCA2 genes⁴. Ovarian carcinoma comprises multiple histopathological subtypes, with epithelial ovarian cancer being the most predominant. This variant originates from the surface epithelial cells that form the ovary's outer protective layer⁵. In 2020, an estimated 313,959 new cases of ovarian cancer were diagnosed globally, resulting in over 207,252 reported deaths⁶. The highest incidence of OC was reported in Brunei, while the highest mortality rate was reported in Samoa in 2020⁷. According to recent epidemiological data, approximately 21,410 new diagnoses and 13,770 ovarian cancer-related deaths were reported in the United States in 2023⁸.

Despite extensive research efforts, robust diagnostic approaches capable of identifying cancer at an early stage and suitable for screening purposes remain unavailable. Besides, there might not be any preliminary symptoms or signs of OC. Consequently, OC is often identified at a late stage, after the tumor has metastasized. Treatment strategies vary based on the stage of the disorder and may include surgery, hormone therapy, and radiation. Therefore, accurate prognosis is essential, particularly for malignant cancers like ovarian cancer, as it supports clinicians in selecting appropriate therapeutic approaches⁹. Moreover, predicting survival outcomes can reduce patient anxiety and enhance their understanding of available therapeutic pathways.

In recent years, the advancement and integration of artificial intelligence (AI) within the medical domain have significantly transformed conventional clinical methodologies, driven by the rapid progression of scientific and technological innovations^{10,11}. AI builds predictive models to analyze vast clinical data and reveal underlying medical insights. At present, AI offers efficient computational approaches for the diagnosis of diverse cancer types through the implementation of ML and DL algorithms^{12,13}. Deep learning enhances medical diagnostics by improving efficiency, minimizing manual effort, and revealing critical features invisible to the human eye, thereby surpassing the accuracy of conventional techniques. It autonomously detects subtle pathological patterns in histopathology images.

Recent studies (2023–2025) have demonstrated the effectiveness of advanced deep learning architectures, including EfficientNet, Vision Transformers (ViTs), and hybrid CNN–Transformer models, for medical image analysis. EfficientNet-based models have shown superior performance in feature extraction (FE) with optimized parameter scaling, while ViTs have proven effective in capturing global contextual information from histopathological images (H-imgs). Furthermore, hybrid CNN–Transformer frameworks have emerged as a promising direction by combining local feature extraction with global attention mechanisms, achieving improved diagnostic accuracy across various medical imaging tasks.^{14–17}

However, several critical challenges remain unaddressed in existing studies. First, most deep learning models suffer from a lack of interpretability, limiting their adoption in clinical decision-making. Second, overfitting remains a significant concern due to limited and imbalanced medical datasets. Third, high computational complexity and training costs restrict the practical deployment of such models in real-world healthcare environments. These limitations highlight the need for an efficient, interpretable, and computationally feasible framework for ovarian cancer classification.

To overcome these limitations, this work incorporates SHAP-based feature selection, which enhances model interpretability by quantifying feature importance and eliminating redundant or less informative features. This not only improves model transparency but also alleviates overfitting and computational overhead, leading to a more robust and efficient classification system.

This study proposed an automated and intelligent framework for the detection and classification of OC using histopathological images. The proposed approach overcomes the limitations of conventional methods by utilizing DL combined with attention mechanisms, which facilitates more accurate and efficient diagnostic prediction.

The proposed model initiates with a preprocessing phase involving bilateral filtering, histogram equalization, and color jittering to improve both the quality and variability of the training dataset. Feature extraction is subsequently performed using the AttResNet50 architecture, a CNN consisting of 50 layers augmented with an attention mechanism, to capture discriminative patterns from H-imgs. To improve the robustness and explainability of the model, SHAP is incorporated to eliminate irrelevant features. Finally, the AttCNN technique is employed to optimize model training and enable effective classification of the dataset. The main contributions of this work are presented as follows:

1. The data augmentation technique is employed to mitigate the problem of limited sample size.
2. The suggested framework uses AttResNet50, a 50-layer attention-based CNN, to pull out important features from H-imgs.
3. The SHAP technique is incorporated to eliminate redundant features during the feature selection phase.
4. We introduce a novel methodology to evaluate the effectiveness of integrating attention mechanisms with convolutional architectures for ovarian cancer image classification.
5. The proposed AttCNN model demonstrates high diagnostic performance, achieving an accuracy of up to 99.09% in detecting ovarian cancer from histopathology images.

We organize the structure of this paper as follows: Sect. [Related works](#) presents a comprehensive review of related work. Section [Methodology](#) describes how the experiments were conducted, covering details about the dataset, the steps taken to prepare the data, how features were extracted and selected, and the design of the proposed AttCNN model. The experimental results are reported in Sect. [Results](#). Lastly, Sect. 5 concludes the study with future recommendations.

Related works

Numerous studies have been carried out to predict cancer outcomes^{18,19}. Some of these investigations have employed analytical techniques and deep neural network models to estimate the outcome of OC, as detailed in Table 1.

In recent years, extensive research efforts have been directed toward enhancing cancer identification through tissue-based imaging and clinical biomarkers such as CA-125 and HE-4. However, these assessment modalities frequently exhibit insufficient sensitivity for dependable early-stage identification of OC²⁷. Additionally, imaging modalities such as magnetic resonance imaging (MRI), computed tomography (CT), and ultrasound are frequently tedious and expensive.

Table 1 Comparative summary of ovarian cancer identification methods.

Ref.	Methodology	Dataset	Performance evaluation
20	Asymmetry and Pearson correlation-based features with KNN, RF, SVM, AdaBoost, DT, XGBoost.	Monitoring, epidemiological, and outcome data (MESOD)	RF: 88.72% accuracy, 82.38% AUC; XGBoost: 20.61% RMSE, 0.4667 R^2 .
21	CNN with convolutional autoencoder for multi-class classification.	1,613 sonographic images	Two classes: 97.2% accuracy, 0.9936 AUC; Five classes: 90.12% accuracy.
22	Three experimental setups with data splitting and 5-fold CV.	288 H&E slides and 175 TMA samples	Best model: 88.2% (CV), 77.5% (TMA).
23	DenseNet-121 with SGD optimizer.	575,930 ultrasound images; validation sets: 8,416 and 6,510 images	AUC: 0.911 (internal), 0.870 and 0.831 (external).
24	CNN with ReLU and Adam optimizer.	11,040 histology images	994.43% accuracy, 95.02% recall, 93.16% specificity.
25	DCNN-based automated ultrasound image analysis for OC detection.	Multicentre ultrasound dataset	AUC: 0.911 (internal), 0.870 & 0.831 (external).
26	CNN-CAE with DenseNet (121/161) for ultrasound image classification and feature refinement.	1,613 ultrasound images	97.2% accuracy, 0.9936 AUC (binary); 90.12% accuracy, 0.9406 AUC (multi-class).

A wide range of computational approaches have been introduced to derive distinctive features through machine learning methodologies for the classification of OC. Some methods utilize wavelet transforms and multi-scale analysis to gather important features, and then they classify the tumors using SVMs and neural networks.

El-Bendary et al.^{[28](#)} introduced a method that uses a boosting strategy along with an ensemble SVM approach to classify ovarian cancer. In^{[28](#)}, the authors employed the T-POT streamlining framework along with the KNN algorithm to identify ovarian cancer. Their model distinguished between healthy, cancerous, intermediate, and non-cancerous categories by analyzing SHG images. Similarly, Jamshid et al.^{[20](#)} utilized skewness and Pearson’s correlation metric to assess the relevance of features derived from the SEER dataset. Subsequently, six ML algorithms, including KNN, SVM, DT, RF, AdaBoost, and XGBoost were implemented for classification tasks. To enhance prediction accuracy, interpretable techniques and SHAP were employed to determine the contribution of each sample to the final prediction.

Yuyeon et al.^{[22](#)} employed a CNN-CAE architecture designed to classify ovarian tumors into five distinct categories while reducing irrelevant data from ultrasound images. The model achieved an accuracy of 96.0% in identifying benign ovarian tumors and 86.85% in differentiating between various tumor types. Ching-Wei et al.^{[22](#)} developed a predictive model to effectively identify OC patients based on H-imgs, utilizing two separate datasets. The study involved two experimental setups: in the first, the dataset was split into 65% for training and 35% for testing; in the second, a 5-fold cross-validation approach was implemented.

Yue et al.^{[23](#)} used a complex neural network comprising 121 layers to differentiate between benign and malignant ovarian tumors using ultrasound images. In this study, three distinct datasets were utilized to assess the model’s performance. The training dataset consisted of 575,930 ultrasound images from 105,532 individuals, among whom 3,755 had OC and 101,777 did not. The internal validation dataset was formed from 8,416 ultrasound images gathered from 868 subjects (266 with ovarian cancer and 602 without). Lastly, the external validation dataset included 6,510 images from 889 cases, comprising 1,253 benign and 5,257 malignant ovarian tissue samples.

In^{[24](#)}, the researchers improved a histopathology image dataset using data augmentation and then divided it into training and validation parts to train a CNN model. The model reached 94% accuracy, correctly detecting 95.12% of cancer cases and 93.02% of healthy ones. In 2025, Sanat Jain et al.^{[29,30](#)} employed deep convolutional architectures, including VGG-19, AlexNet, ResNet, and GoogLeNet, for feature representation. The extracted features were subsequently classified using a Support Vector Machine (SVM) classifier.

Xu et al.¹⁴ demonstrated the effectiveness of Vision Transformers (ViTs) for histopathological image analysis; however, their work was limited to standalone ViT architectures, lacking hybrid integration and interpretability mechanisms. Xu et al.¹⁵ reviewed medical image segmentation techniques, including hybrid models, though without specific emphasis on histopathological data and practical clinical deployment. Similarly, Kim et al.¹⁶ explored hybrid deep learning approaches in radiology imaging; however, their study was confined to radiological modalities and did not address dataset variability or computational efficiency. In contrast, Md. Easin et al.¹⁷ proposed a hybrid ViT–CNN framework combining DeiT-Small and EfficientNet-B7 for classifying Alzheimer’s disease-related cell types from microscopy images, achieving superior performance compared to conventional CNNs and vision–language models. Nevertheless, their approach involves high computational cost, lacks explainability, and is validated on limited domain-specific datasets, which may introduce bias and restrict generalizability^{31,32}. Overall, despite these advancements, the application of such models to histopathological ovarian cancer classification remains limited, particularly with respect to interpretability, overfitting mitigation, and computational efficiency.

In recent years, a variety of methods have been implemented to incorporate DL techniques for the early detection of OC. By employing data sources such as gene expression profiles, histopathological slides, and ultrasound images, these studies suggest that deep learning is effective in the identification and classification of various forms of ovarian cancer. These techniques have exhibited substantial enhancements in the identification of molecular subtypes and diagnostic accuracy.

Methodology

The data collection procedure, image pre-processing procedures, feature extraction and selection methods, and the suggested framework are all described in this part. Figure 1 illustrates the schematic representation of our proposed approach. Additionally, we incorporated Algorithm 1 to illustrate the pseudocode of our suggested method for OC classification.

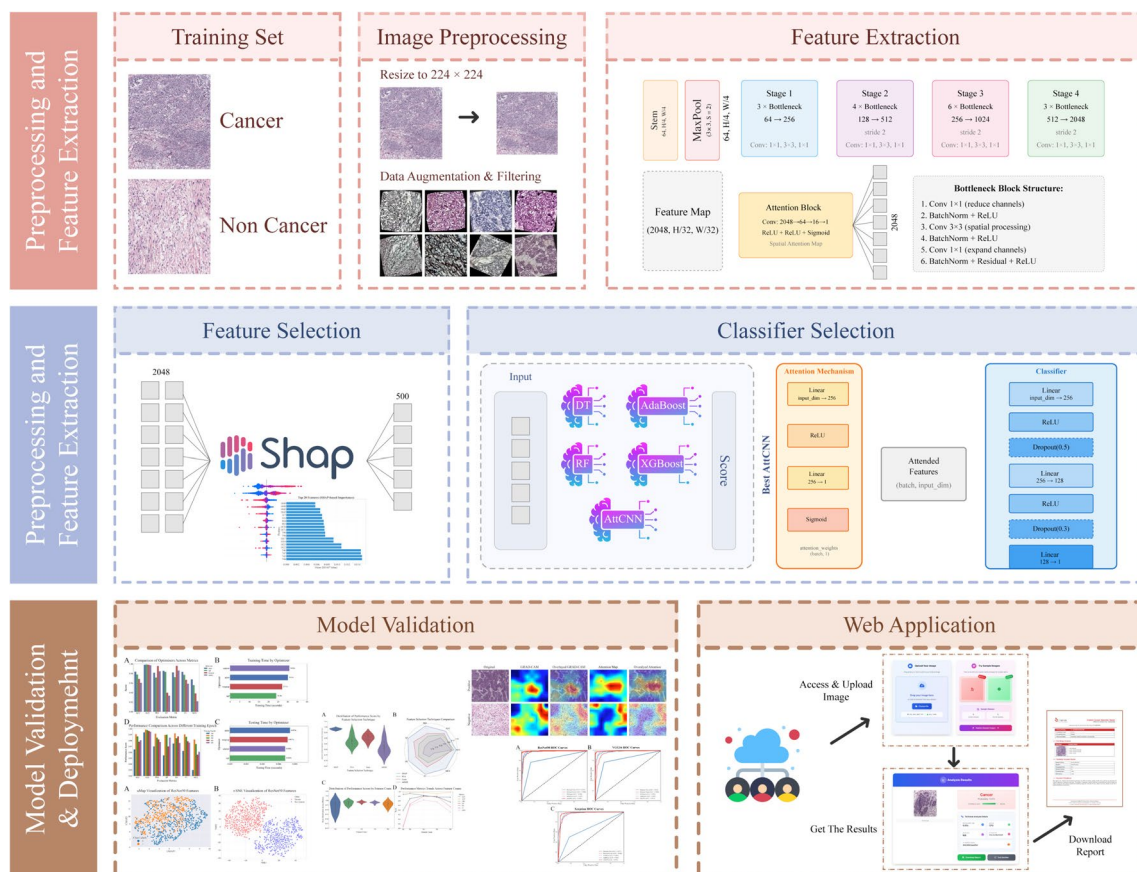


Fig. 1 The flowchart of our proposed methodology.

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1:  $E_p \leftarrow$  Number of Epochs
2:  $W \leftarrow$  Transfer Learning Model Parameter
3:  $\eta \leftarrow$  Learning Rate
4:  $b_s \leftarrow$  Batch Size
5:  $D \leftarrow$  Ovarian Cancer Dataset
6: Output: The assessment metrics on the test dataset.
7: Dataset Preprocessing:
8:  $X_{train} \leftarrow preprocessing(D)$ 
9:  $X_{test} \leftarrow preprocessing(D)$ 
10: Apply filtering (bilateral, CLAHE, histogram equalization, Gaussian blur)
11: Apply augmentation (flip, rotation, affine, gamma, sharpening, color jitter)
12: Resize:  $216 \times 250 \rightarrow 224 \times 224$ 
13: Feature Extraction using AttResNet50:
14: for local epoch  $e_p \leftarrow$  from 1 to  $E_p$  do
15:   for  $b_i = (x_s, y_s) \in$  random batch from  $X_{train}$  do
16:     Optimise model parameters
17:      $W_e \leftarrow W_e - \eta(\Delta(\mathcal{L}(W_e; b_s)))$ 
18:      $f_{train} \leftarrow ComputeFeatures(W_e, X_{train}, 2048)$ 
19:   end for
20: end for
21: Feature Selection using SHAP:
22: Compute  $\phi_i(f, x) = \sum_{z' \subseteq x} \frac{|z'|!(|m|-|z'|)!}{|m|!} [f_x(z') - f_x(z' \setminus i)]$ 
23:  $f_{best} \leftarrow$  Select top ranked features
24: Ovarian Cancer Classification using AttCNN:
25: for epoch  $e_p \leftarrow$  from 1 to  $E_p$  do
26:   for  $b_i = (x_s, y_s) \in$  random batch from  $f_{best}$  do
27:     CNN Block: Conv2D + BatchNorm + ReLU + MaxPool
28:     Attention Block:  $u_i \leftarrow \tanh(W_i h_i + b_i)$ 
29:      $\alpha_i \leftarrow \frac{\exp(u_i^T u_w)}{\sum_j \exp(u_j^T u_w)}$ 
30:      $h_i \leftarrow \sum_j \alpha_j h_j$ 
31:     Output Block: FC layers with Dropout
32:      $\mathcal{L} = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$ 
33:   end for
34: end for
35:  $TrainedModel \leftarrow AttCNN(f_{best}, y_{train})$ 
36:  $Pred \leftarrow TrainedModel(X_{test})$ 
37:  $Evaluation\ metrics \leftarrow ComputeMetrics(Pred, y_{test})$ 

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Algorithm 1 Proposed Algorithm for Ovarian Cancer Classification

Dataset

In this study, we have utilized the OC histopathological image dataset³³ to carry out analyses with our proposed framework. This dataset was sourced from the tissue repository of the prestigious Tri-Service General Hospital and the National Defense Medical Center in Taipei, Taiwan. It comprises 987 high-resolution digital H-imgs, each with dimensions of 216×250 pixels across three spectral channels (RGB), forming an array structure of (987, 216, 250, 3). The average physical dimensions of these images are approximately 27.34×20.66 mm. Each image is annotated with a corresponding label, categorizing it into one of two distinct classes: OC or non-OC. Specifically, the dataset consists of 481 OC cases and 506 non-OC cases. For the purpose of model training and evaluation, the dataset was systematically stratified using the holdout validation approach, maintaining a 70:15:15 split. This partitioning resulted in a training collection of 790 images, a validation collection of 169 images, and a test set of 169 images, ensuring a robust assessment of our methodology. In addition, a 5-fold cross-validation procedure was conducted to further validate the robustness and generalizability of the proposed approach.

Data preprocessing

Before investigations are conducted, specific techniques are employed to preprocess data in order to improve its integrity and achieve improved results. In this study, we implemented a comprehensive preprocessing pipeline, which included filtering techniques, data augmentation, image scaling, and normalization. Initially, various filtering methods were applied to the raw dataset, including bilateral filtering, CLAHE, histogram equalization, and Gaussian blur, to improve image clarity and contrast. Subsequently, data augmentation strategies were employed to infuse variability and optimize the model's generalization ability. These transformations comprised random horizontal flipping, random rotation (20 degrees), random affine transformation, gamma correction, sharpening, and color jittering (brightness 0.2, contrast 0.2, saturation 0.2, hue 0.1). To ensure consistency in input dimensions, all preprocessed images were resized from 216×250 to 224×224 and converted into tensors before feeding them into feature encoding module. Normalization was applied using channel-wise mean values of (0.485, 0.456, 0.406) and standard deviations of (0.229, 0.224, 0.225), following the standard ImageNet-based z-score normalization scheme. Finally, pixel intensity values were rescaled to the (0,1) range by dividing by 255, further improving computational efficiency and stability during convergence. A subset of the filtered images is shown in Fig. 2.

Feature extraction using AttResNet50

The process of feature extraction enables us to retrieve significant attributes from input images and is vital for any classification task. We have developed an attention-based ResNet50 model, termed AttResNet50, as our feature extractor to enhance deep feature extraction capabilities. Typically, ResNet-50³⁴ is a pre-trained architecture that can identify different images in 1,000 categories. It was trained using the ImageNet dataset. The residual layers in ResNet-50 play a crucial role in propagating substantial gradient values to their preceding adjacent layers. This layer enables the model to efficiently extract intricate and meaningful patterns while alleviating the vanishing gradient problem^{35,36}. In our study, ResNet50 comprises 49 convolutional layers and a single fully connected (FC) layer³⁷. Together, these convolutional regions quickly downsample the most relevant information from the raw input, discover local patterns, and capture sophisticated spatial representations and distinguishing characteristics. This is known as the “feature extraction” step. In the subsequent phase, the FC layers, also referred to as

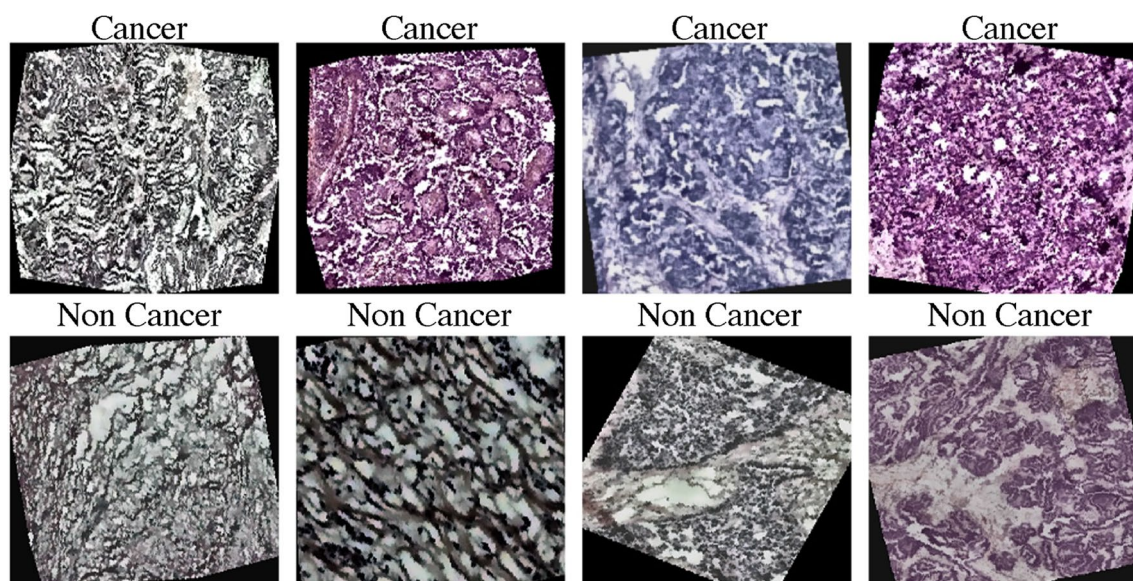


Fig. 2 Visualizing the impact of various filtering techniques on the original input image.

the “classifier,” utilize the extracted features as input to carry out identification tasks by implementing activation schemes and learned weight parameters. In our approach, we have modified this architecture to enable it to perform both feature extraction and classification concurrently, allowing for the extraction of 2048 features and the computation of classification loss in a single forward pass. Specifically, the feature extraction component of the original model remains unchanged, while the classification layer is augmented with an attention mechanism. The model architecture is illustrated in the Fig. 3.

The AttResNet50 model takes a standard 224×224 RGB image as input and generates feature maps of dimensions $2048 \times 7 \times 7$ during the feature extraction phase, which are subsequently reduced via global average pooling to a compact 2048-dimensional feature vector, enabling efficient and discriminative representation for classification. These maps of features are then squashed into a one-dimensional vector of size 1×100352 before to being processed by the attention block. Within the attention block, a series of 1×1 convolutional operations is applied to generate a spatial attention map. The attention-weighted features undergo an adaptive average pooling operation, resulting in a one-dimensional feature vector of size 1×2048 , which contributes directly to deep FE. Our dataset has two distinct groups, OC and non-OC, thus we set the output dimension of the FC layer to 1×2 . We implemented a forward hook³⁸ at the FC layer in our framework to dynamically extract 2048 features. We use the Grid Search method for hyperparameter tuning. For optimization, ReLU was employed as the activation function and Adam as the optimizer, with a learning rate of 0.001, a weight decay parameter of 0.0005, and a momentum factor of 0.9. Additionally, Cross-Entropy loss was utilized as the loss function. To mitigate overfitting, dropout rates of 0.5 and 0.25 were applied to the FC layer. This optimized scheme was trained using hyperparameter modifications for 100 runs, preserving the smallest validation loss model weights for future usage.

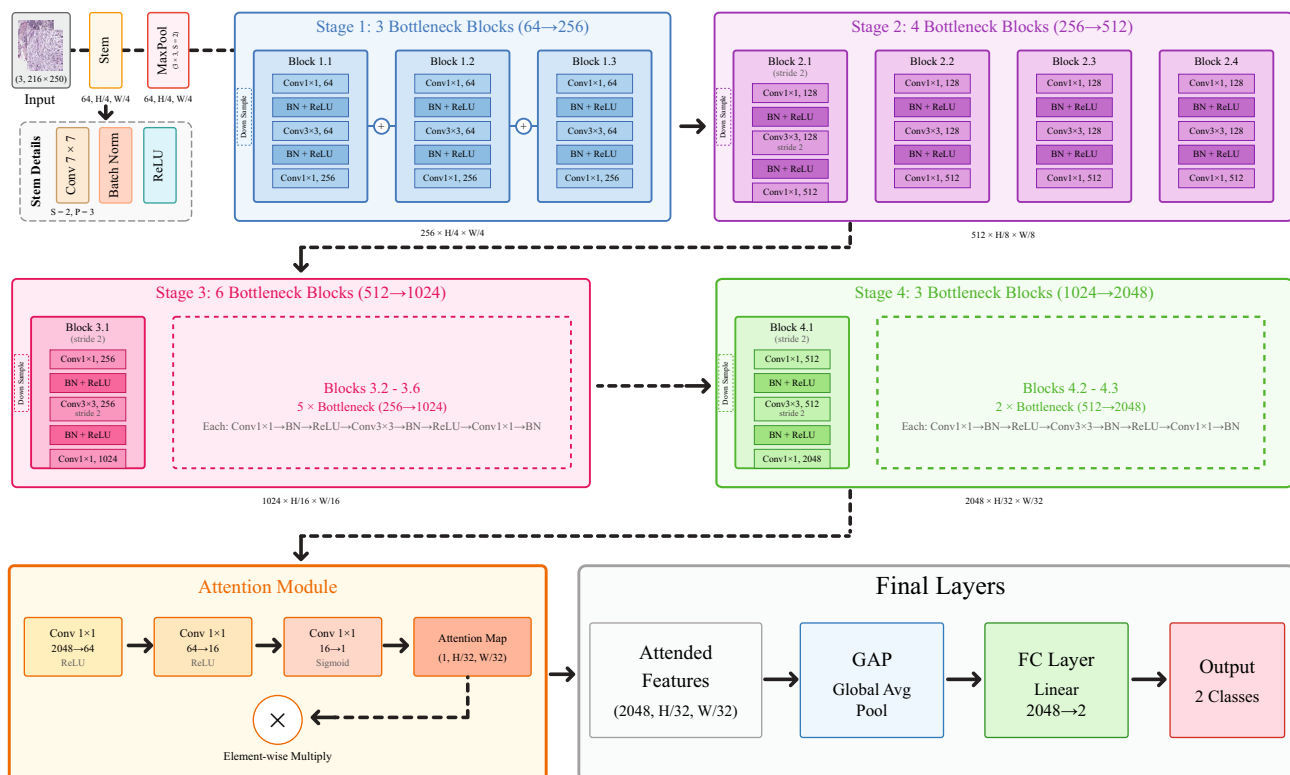


Fig. 3 Architecture of the proposed AttResNet50 feature extraction scheme.

Feature selection using SHAP

Feature selection (FS) has been a mainstay in ML for determining which instance properties are the most important and informative for improving model performance. SHAP³⁹ was leveraged, through which Shapley values derived from cooperative or coalitional game theory were computed⁴⁰ and significance scores were assigned to individual features for a particular prediction. Traditionally, in the context of a coalitional game involving a group of participants $N = \{1, 2, \dots, n\}$, the Shapley score associated with a participant i , represented as ϕ_i , is mathematically expressed as follows:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)] \quad (1)$$

In Eq. 1, N indicates the total number of participants, S signifies a subset of participants where $S \subseteq N \setminus \{i\}$, encompassing all feasible subsets for participant i . The expression $(v(S \cup \{i\}) - v(S))$ measures the incremental contribution, where $v(S \cup \{i\})$ and $v(S)$ correspond to the coalition's value with and without player i , respectively.

In this context, a game is formulated as a predictive modeling task $f(\cdot)$ for an individual instance x , where the participants indicate the features of the sample, expressed as $x = x_1, x_2, \dots, x_n$. Each feature contributes to the overall outcome, while the payoff is represented by the predicted output $f(x)$. SHAP employs an additive feature contribution technique that emphasizes the “instance-level method” to elucidate the model output $f(\cdot)$ by utilizing a surrogate model $g(\cdot)$, formulated as a linear combination of binary variables:

$$g(x') = \phi_0 + \sum_{i=1}^M \phi_i x'_i \quad (2)$$

where x' denotes a simplified representation of the input which is transformed into the original input using a mapping function $x = h_x(x')$, potentially retaining less information than x . The variable x'_i indicates whether the i -th feature is present ($x'_i = 1$) or absent ($x'_i = 0$) and m represents the number of streamlined attributes. Additionally, $\phi_i \in R$ represents the attribution value assigned to the i -th feature. This approach utilizes Eq. 1 to efficiently calculate the attribute importance score for the predictive model $f(\cdot)$ and the corresponding instance x as follows:

$$\phi_i(f, x) = \sum_{z' \subseteq x'} \frac{|z'|! (|m| - |z'| - 1)!}{|m|!} [f_x(z') - f_x(z' \setminus i)] \quad (3)$$

where $|z'|$ signifies total count of non-zero elements in z' , and $z' \subseteq x'$ denotes all possible vectors z' where the non-zero entries form a subset of those in x' . Eq. 3 defines the coalition-based game theory for feature contribution, as formulated in Eq. 1, ultimately yielding the Shapley score. The SHAP approach utilizes the initial model and instances as input to calculate the Shapley scores for each sample, which can be used to assess feature significance in explainable FS techniques.

We applied the XGBoost model as our predictive algorithm and computed the importance score for each attribute. Based on these scores, the features were ranked from largest to smallest, and a subset of the most significant features was selected for training using an ML technique.

Proposed model architecture

In order to the early detection of OC, we developed an attention-based CNN model named AttCNN, which enables the model to emphasize on the most significant features for classification. Our model architecture comprises three components: the CNN block, the attention block, and the output block.

CNN block

CNN is widely utilized for computer vision, natural language processing, and personalized recommendation systems⁴¹. The first component of our model is the CNN block, which comprises multiple convolutional layers that employ filters on the input image, extracting and analyzing local patterns to generate feature maps⁴². In our model, the CNN block starts with an input image of size $224 \times 224 \times 3$, which passes through three consecutive convolutional layers, each utilizing 3×3 kernels and producing feature maps with 32, 64, and 128 output channels, respectively. Each convolutional layer is followed by a batch normalization³⁶ and a ReLU activation function⁴³, which introduces non-linearity and enhances the model's ability to learn complex feature representations. The convolutional kernels traverse the input image to generate feature maps that extract distinct spatial patterns and structural attributes within the data. The max pooling phase has a pooling window of 2×2 with a stride of 2, while the stride dimension is set to (1,1) with same padding. To expedite quicker convergence, the ReLU (Rectified Linear Unit) activation function is employed. ReLU can be expressed as follows:

$$\text{ReLU}(x) = \begin{cases} x, & \text{if } x > 0 \\ 0, & \text{if } x \leq 0 \end{cases} \quad (4)$$

Here, the ReLU function illustrates that when the input x is less than or equal to 0, the output remains zero, whereas for positive values of x , the output corresponds directly to the input. Subsequently, the model incorporates a series of processing blocks meticulously designed to extract hierarchical representations while mitigating the vanishing gradient problem. Then the resulting feature maps are flattened, transforming them into a vector that serves as input for the attention mechanism. Furthermore, the multi-class cross-entropy loss function was used, and the Adam algorithm was implemented with a learning frequency of 0.001.

Attention block

The attention mechanism, regarded as the most crucial portion of our proposed model, is utilized to focus on informative features and suppress irrelevant information⁴⁴. Within this framework, the attention mechanism employs a content-based soft attention approach that dynamically assigns importance weights to different spatial locations in the feature representation⁴⁵. Specifically, the flattened feature vector of dimension $128 \times 28 \times 28$ undergoes a linear transformation, followed by ReLU activation which results in a hidden dimension of 256. This intermediate representation is subsequently processed through an additional linear layer that produces a scalar value for each spatial location. These attention scores are normalized using a sigmoid activation function, resulting in attention weights constrained within the range $[0, 1]$. The computed attention weights optimize the feature representation by emphasizing salient regions while reducing the influence of irrelevant ones during inference. Furthermore, the precise implementation of the attention mechanism utilized in this study is mathematically represented by the following equation:

$$u_i = \tanh(W_i h_i + b_i) \quad (5)$$

$$\alpha_i = \frac{\exp(u_i^T u_w)}{\sum_i \exp(u_i^T u_w)} \quad (6)$$

$$h_i = \sum_i \alpha_i h_i \quad (7)$$

where h_i denotes the hidden state of the encoder at the i -th time step, W_i represents the learnable weight matrix associated with the transformation, b_i denotes the corresponding bias vector, u_i is the transformed hidden representation obtained after applying the hyperbolic tangent activation function, and α_i signifies the attention weight for the i -th position.

Output block

Finally, the output block processes the attention block's output to generate the final classification result. The output block consists of a fully connected neural network (FCNN) that includes a sequence of linear layers, ReLU activations, and dropout layers to reduce overfitting. The input is mapped to 256 concealed units by the initial linear layer, accompanied by a ReLU activation with a dropout rate of 0.5. Another linear transformation reduces the dimensions from 256 to 128, with an additional dropout rate of 0.3 applied. The last layer output $\hat{y}$ is the predicted value of the model. As a binary classification task, we utilize cross-entropy loss to train our model:

$$\text{loss} = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (8)$$

where y represents the ground truth label, $\hat{y}$ denotes the predicted probability that the label is 1, $\log(\hat{y})$ measures the error when the true label is 1, and $\log(1 - \hat{y})$ measures the error when the true label is 0.

Results

Evaluation metrics

We utilized various performance metrics to assess the effectiveness of our proposed model at each stage after feature extraction. Accuracy, specificity, sensitivity, Matthew's correlation coefficient (MCC), and precision were considered as evaluation metrics. Mathematically, these metrics are defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TN + TP + FN + FP} \quad (9)$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (10)$$

$$\text{Specificity} = \frac{TN}{FP + TN} \quad (11)$$

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TN + FP)(TN + FN)(TP + FN)}} \quad (12)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (13)$$

$$\text{F1 score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (14)$$

Where the total number of positive instances that are accurately predicted is denoted by TP (true positive), TN (true negative) indicates the total number of correctly recognized negative instances, FP (false positive) denotes the number of negative instances erroneously identified as positive, and FN (false negative) is the count of positive instances misclassified as negative. Additionally, the overall performance of our proposed model and other ML classifiers was quantitatively assessed using the Area Under the Curve (AUC) metric.

All experiments were conducted in a macOS Sonoma 14.6.1 environment utilizing the Apple M2 chip architecture with an 8-core CPU, 10-core GPU, 16GB of unified memory, 256GB SSD storage, and a 16-core Neural Engine delivering 100GB/s memory bandwidth. Python 3.11, along with Pandas, PyTorch 2.6.0, NumPy, Matplotlib, and other computational packages, was employed to develop and evaluate our proposed methodology. The source codes employed in this work will be available on GitHub at: <https://github.com/camlas/ovarian-cancer-histopathology>

Table 2 Experimental results of feature extractor performance with various classifiers.

Feature extractor	Classifier	ACC	AUC	PRE	SP	SN	F1	MCC
AttResNet50	DT	0.8176	0.8165	0.8413	0.7713	0.8617	0.8048	0.6364
	RF	0.9686	0.9956	0.9500	0.9875	0.9506	0.9684	0.9379
	XGBoost	0.9777	0.9976	0.9655	0.9896	0.9664	0.9774	0.9557
	AdaBoost	0.9737	0.9972	0.9541	0.9938	0.9545	0.9735	0.9481
	AttCNN	0.9878	0.9995	0.9835	0.9917	0.9842	0.9876	0.9757
VGG16	DT	0.8105	0.8099	0.8196	0.7838	0.8360	0.8013	0.6210
	RF	0.9108	0.9841	0.9645	0.8482	0.9704	0.9027	0.8268
	XGBoost	0.9534	0.9945	0.9780	0.9252	0.9802	0.9509	0.9079
	AdaBoost	0.9443	0.9934	0.9863	0.8981	0.9881	0.9402	0.8916
	AttCNN	0.9848	0.9973	0.9916	0.9771	0.9921	0.9843	0.9697
Xception	DT	0.6798	0.6766	0.7273	0.5489	0.8043	0.6256	0.3661
	RF	0.9220	0.9823	0.9430	0.8940	0.9486	0.9178	0.8448
	XGBoost	0.9240	0.9829	0.9319	0.9106	0.9368	0.9211	0.8481
	AdaBoost	0.9402	0.9860	0.9360	0.9418	0.9387	0.9389	0.8804
	AttCNN	0.9676	0.9815	0.9610	0.9730	0.9625	0.9669	0.9352

Table 3 Computational complexity comparison of different feature extractor and classifier combinations.

Feature extractor	Classifier	Parameters	GFLOPs	Inference time (ms)	Computational time (s)
AttResNet50	DT	25.6M	4.1	18.9	2.01
	RF	25.6M	4.1	19.4	2.08
	XGBoost	25.6M	4.1	20.3	2.17
	AdaBoost	25.6M	4.1	19.8	2.12
	AttCNN	25.9M	4.3	17.4	1.93
VGG16	DT	138M	15.3	33.1	4.36
	RF	138M	15.3	32.7	4.28
	XGBoost	138M	15.3	33.4	4.41
	AdaBoost	138M	15.3	32.9	4.35
	AttCNN	138.4M	15.6	31.6	4.20
Xception	DT	22.9M	8.4	25.1	3.25
	RF	22.9M	8.4	24.7	3.19
	XGBoost	22.9M	8.4	25.4	3.28
	AdaBoost	22.9M	8.4	24.9	3.23
	AttCNN	23.2M	8.7	23.8	3.10

Feature extractor and classifier selection

To identify the most effective FE, we implemented three different CNN architectures, namely AttResNet50, VGG-16, and Xception. In this case, a transfer learning technique was adopted, wherein pre-trained weights from large-scale benchmark datasets were utilized instead of training the networks from scratch. The original FC layers of these CNN architectures were eliminated and substituted with five unique ML classifiers: DT, RF, XGBoost, AdaBoost, and AttCNN. Each CNN architecture served as a feature extractor, and the extracted representations were independently classified using the corresponding ML algorithms. This combinatorial method evaluates and compares the effectiveness of several feature extraction techniques in association with various ML-based classifiers for the OC histopathological image dataset. The evaluation was conducted on the test subset, and from this analysis, the optimal feature extractor, the most effective classifier, and the best-performing combination were selected based on their predictive accuracy. The detailed results of all evaluated combinations are summarized in Table 2.

Table 3 summarizes the computational complexity analysis of the comparative models and the proposed architecture. The results indicate that AttResNet50+AttCNN has superior classification performance at a moderate

computational cost, as measured by the number of trainable parameters and GFLOPs. Although the models based on VGG16 have a higher number of parameters and computational costs, the proposed model has better performance.

Results in Table 2 show that AttResNet50 and all five classifiers achieved the greatest average accuracy of 94.51% among all examined FEs. In contrast, Xception achieved the lowest average accuracy of 88.67%, while VGG-16 achieved a mean accuracy of 92.08%, which is 2.43% lower than that of ResNet50. Additionally, AttResNet50 attained the highest average AUC score of 96.13%, exceeding Xception and VGG-16 by 3.94% and 0.55%, respectively. This FE has the greatest average values for specificity, MCC, and F1-score, averaging 94.68%, 89.08%, and 94.23%, respectively. The average MCC scores of AttResNet50 are 11.59% and 4.74% greater than those of Xception and VGG-16, respectively. In terms of specificity and F1-score, AttResNet50 outperforms the other two extractors by 9.31%, 6.03% and 6.82%, 2.64%, respectively. When compared with the other feature extractors, these differences are notably substantial, indicating that AttResNet50 achieved superior performance in terms of predictive accuracy and computational efficiency. Therefore, we have selected AttResNet50 as our feature extractor.

From Table 2, it is clear that the AttCNN classifier achieved the highest average accuracy of 98.01%, which is 21.08%, 4.63%, 2.74%, and 2.84% higher than those of DT, RF, AdaBoost, and XGBoost, respectively. The AttCNN model also obtained the maximum average AUC of 99.28%, surpassing the other classifiers mentioned by 22.51%, 0.55%, 0.06%, and 0.11%, respectively. In terms of the MCC, the DT, RF, AdaBoost, and XGBoost classifiers yielded average scores of 54.12%, 86.98%, 90.67%, and 90.39%, respectively, while AttCNN achieved the highest average MCC of 96.02%. Additionally, AttCNN achieved the highest average values for other performance metrics, including specificity and F1-score, which were 98.06% and 97.96%, respectively. For ACC, AUC, MCC, SP, and F1-score, the AdaBoost classifier gets the second-highest average values. These values are 2.74%, 0.06%, 5.35%, 3.6%, and 2.87% less than the AttCNN classifier. Therefore, we concluded that the AttCNN classifier demonstrated superior performance compared to all other classifiers employed in this study on the test dataset. These outcomes help us to choose AttCNN as the most effective classifier. Figure 4 shows the ROC curves corresponding to each feature extractor integrated with the five distinct classifiers. Furthermore, Fig. 5 presents the visualization of the feature space using UMAP and t-SNE, based on features extracted by AttResNet50.

The experimental findings also confirm that the AttCNN classifier integrated with each feature extractor consistently outperformed all other combinations in terms of ACC, AUC, MCC, SP, and F1-score. Conversely, each classifier paired with AttResNet50 yielded superior performance compared to other feature extractors. It is evident that the AttResNet50 and AttCNN classifier combination achieved the highest accuracy of 98.78%, which is substantially higher than all other pairings. Based on this finding, we chose AttResNet50 as the FE and AttCNN as the classifier for our proposed model.

Impact of feature selection techniques

The initial feature subsets extracted from the feature extraction methods may encompass irrelevant or redundant information that can degrade the effectiveness of the classifiers. Therefore, FS methods were utilized to identify the most informative and relevant features from these subsets. In this study, four FS methods, including PCA⁴⁶, LASSO⁴⁷, mRMR⁴⁸, and SHAP³⁹ were utilized to refine the features extracted by AttResNet50. The experimental outcomes of these methods on the OC dataset are shown in Table 2. By comparing Tables 2 and 4, we can see that applying an FS approach to the dataset improved the model's performance. As shown in Table 2, SHAP yielded the highest accuracy 99.09%, which is 3.09%, 5.09%, and 6.09% higher than PCA, LASSO, and mRMR, respectively. However, for the PCA technique, the ACC, AUC, Sen, Spe, MCC and F1 scores were 96.00%, 92.00%, 86.00%, 85.00%, 82.00%, and 85.00%, respectively. In contrast, the AUC, Sen, Spe, MCC, and F1 scores for mRMR were 87.57%, 80.00%, 78.00%, 75.00%, and 78.00%, which are 12.28%, 19.21%, 20.96%, 23.18%, and 21.06% lower than SHAP, respectively. For the LASSO, the AUC, Sen, Spe, MCC, and F1 scores were 94.00%,

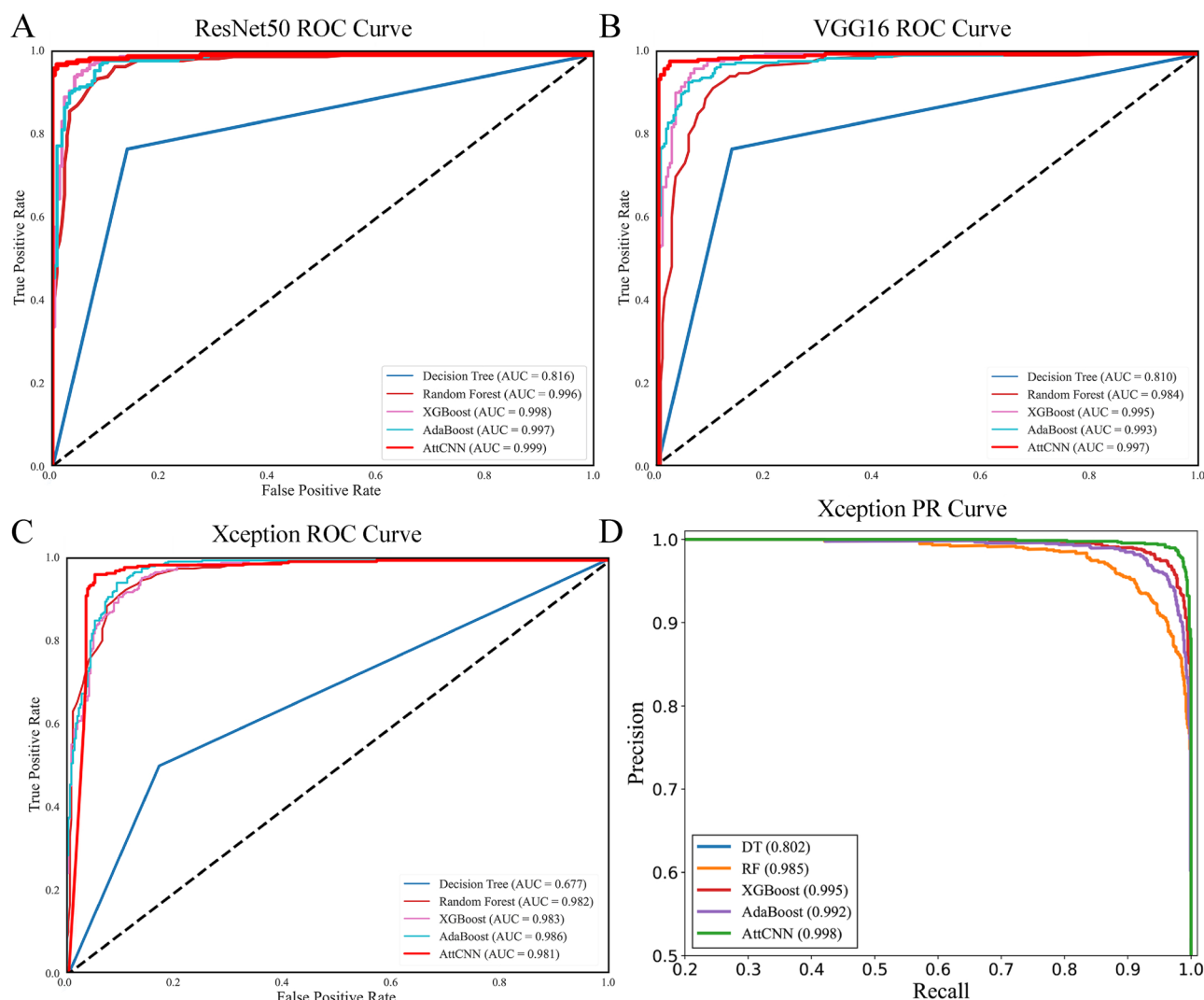


Fig. 4 ROC-AUC evaluation of three FE. (a) AttResNet50; (b) VGG-16; (c) Xception. (d) PR Curve for Xception.

89.00%, 91.56%, 85.00%, and 90.06%. Among all FS techniques, SHAP consistently outperformed others across all evaluation metrics, achieving an accuracy of 99.09%, AUC of 99.85%, sensitivity of 99.21%, specificity of 98.96%, MCC of 98.18%, and an F1-score of 99.06%. These findings indicate the consistent and robust performance of SHAP in selecting predictive features. Confusion matrix of four feature selection technique is visualized in the Supplementary Figure 1.

Additionally, we investigated the impact of different feature subset sizes on model performance. Table 5 presents the evaluation results for subsets comprising the top 200, 300, 500, 700, and 900 features selected using SHAP. From the experimental outcomes, we observe that the ACC, AUC, Sen, Spe, MCC and F1 scores for 200, 300, 500, and 700 features were slightly better than 900 features. The optimal results were obtained for 500 features: the ACC, Sen, Spe, MCC, Pre and F1 scores for this subset were 99.09%, 99.21%, 98.96%, 98.18%, 99.17% and 99.06%, respectively. Notably, the lowest results were obtained using the 900-feature subset. The ACC, Sen, Spe, MCC, Pre, and F1 scores for this subset were 97.77%, 98.81%, 96.67%, 95.56%, 98.73%, and 97.69%, respectively, which are 1.32%, 0.40%, 2.29%, 2.62%, 0.44%, and 1.37% lower than those obtained with the 500-feature subset. Thus, the optimal subset of 500 features identified through the SHAP technique was used to develop our proposed model. We have also presented a violin plot and a radar chart for our chosen feature

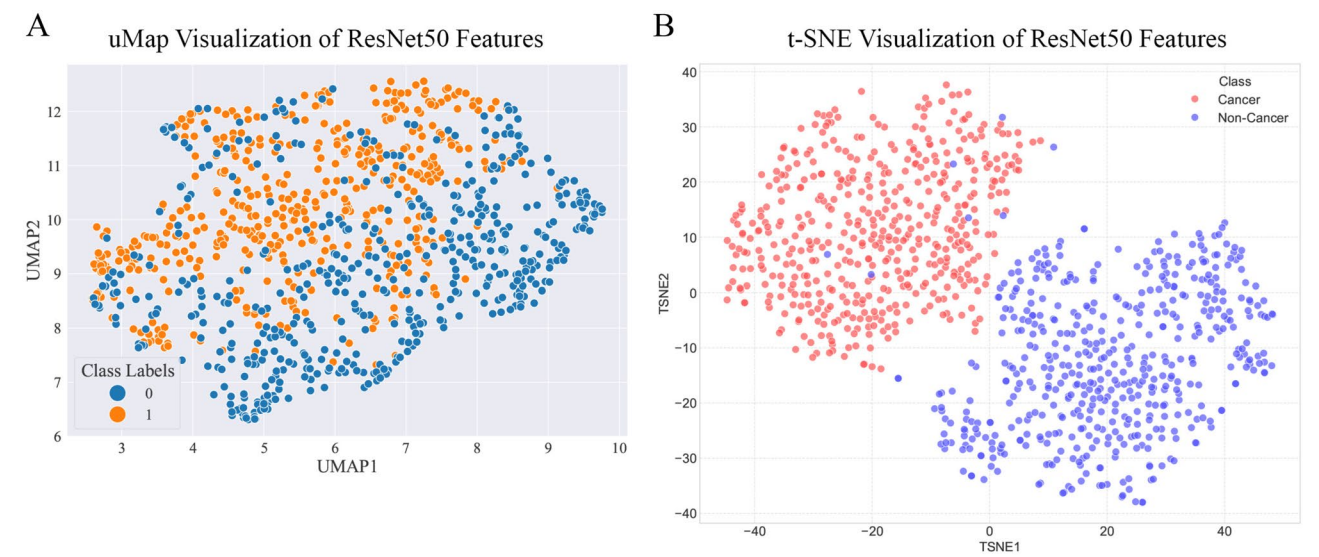


Fig. 5 Feature space visualization using UMAP and t-SNE reduction techniques applied to AttResNet50-extracted features.

Table 4 Experimental results obtained from applying various feature selection techniques.

Technique	ACC	AUC	PRE	SP	SN	F1	MCC
SHAP	0.9909	0.9985	0.9917	0.9896	0.9921	0.9906	0.9818
PCA	0.9600	0.9200	0.8600	0.8500	0.8600	0.8500	0.8200
Lasso	0.9400	0.9400	0.8900	0.9156	0.8900	0.9006	0.8500
mRMR	0.9300	0.8757	0.7900	0.7800	0.8000	0.7800	0.7500

Table 5 Prediction performance using SHAP-based feature selection with different feature combinations in the proposed model.

No. of feat.	ACC	AUC	PRE	SP	SN	F1	MCC
200	0.9605	0.9970	0.9933	0.9252	0.9941	0.9580	0.9229
300	0.9848	0.9982	0.9957	0.9730	0.9960	0.9842	0.9698
500	0.9909	0.9985	0.9917	0.9896	0.9921	0.9906	0.9818
700	0.9889	0.9955	0.9876	0.9896	0.9881	0.9886	0.9777
900	0.9777	0.9987	0.9873	0.9667	0.9881	0.9769	0.9556

selector that is shown in Fig. 6. Furthermore, we applied a Grad-CAM⁴⁹ visualization utilizing our selected feature extractor (AttResNet50) to provide an interpretable visual rationale. Grad-CAM enhances interpretability by highlighting the most influential regions within an image that contribute to the CNN’s decision-making process. It produces a heatmap by leveraging the feature maps from the final convolutional layer, thereby emphasizing the most salient visual features relevant to classification. Subsequently, we also applied attention-based visualization techniques. Compared to conventional Grad-CAM heatmaps, attention maps demonstrated improved localization and sharper delineation of discriminative regions within the H-imgs. Figure 7 displays the original images in conjunction with their respective Grad-CAM renderings for each category.

Main classifier parameter, optimizer and ablation study

Effective hyperparameter tuning plays a pivotal role in optimizing the performance and training efficiency of DL-based predictive models. These parameters directly influence both convergence behavior and classification accuracy. In the case of our proposed AttCNN model, the major hyperparameters include learning rate, batch size,



Fig. 6 Violin plots and radar charts depicting the experimental findings of four different FS techniques (A,B), along with the corresponding feature counts (C,D).

dropout rate, loss function, and the number of training epochs. In this study, we adopted a learning rate of 0.001, a batch size of 32, and a training duration of 200 epochs. Here, the Adam optimizer was employed to minimize the loss during training. Additionally, the BCEWithLogitsLoss function was utilized within the hidden layers. Table 6 presents a detailed overview of the hyperparameter settings for the proposed AttCNN model.

The choice of optimization algorithm is another crucial determinant of model performance, as it governs the rate and stability of convergence by generating gradients and iteratively minimizing the loss function. In our study, we evaluated four common optimizers: SGD, Adam, RMSProp, and AdamW to find the best one for the AttCNN architecture. The comparative results of these algorithms are shown in Table 7. The results indicate that the Adam optimizer attained an AUC of 99.85%, SGD yielded 99.88%, RMSProp achieved 99.83%, and AdamW scored 99.72%. Correspondingly, the ACC values achieved by SGD, Adam, RMSProp, and AdamW were 99.09%, 97.77%, 96.86%, and 96.05%, respectively. As shown in Table 7, it can be inferred that Adam and RMSProp are more favorable in terms of producing lower loss values and reduced execution times compared to SGD. In binary classification, the loss values obtained using SGD, Adam, RMSProp, and AdamW were 0.0417, 0.1117, 0.0176, and 0.0238, respectively, while their corresponding execution times were 30.1001s, 24.0389s, 27.1115s, and 30.7972s. These findings suggest that Adam provides greater computational efficiency and achieves lower

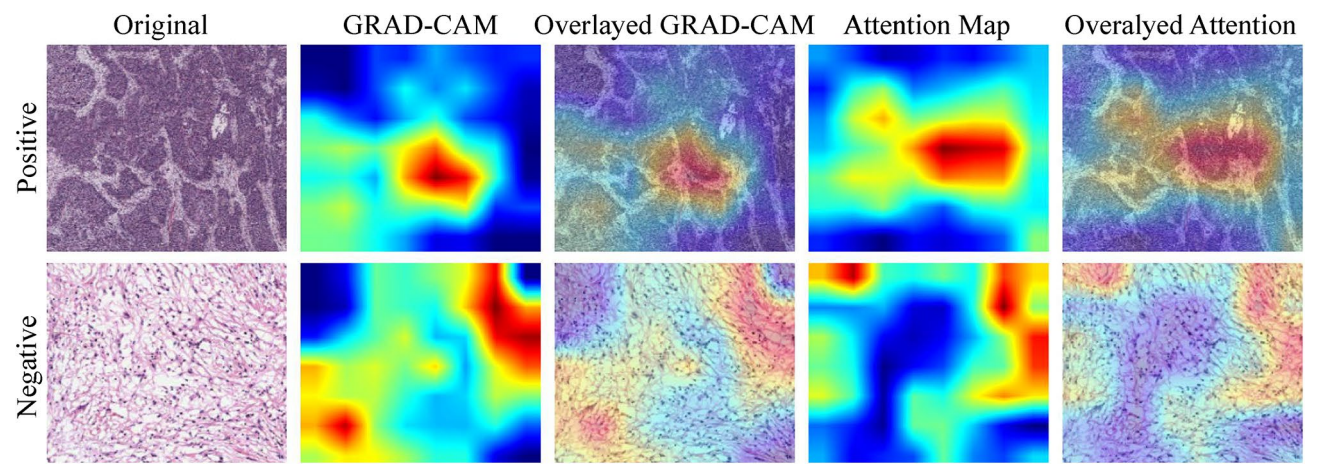


Fig. 7 Grad-CAM visualizations are generated from the final convolutional layer of the AttResNet50 feature extractor, where highly influential locations are showed in red and less relevant areas in blue, which helps in interpreting feature relevance.

Table 6 Hyperparameter settings of our proposed model.

Name	Parameter
Input dimension	2048
Hidden dimension	256
Attention layer dimensions	[2048, 256, 1]
Attention layer activation functions	[ReLU, Sigmoid]
Classifier layer dimensions	1
Classifier layer activation functions	[ReLU, ReLU, None]
Dropout rates	[0.5, 0.3]
Loss function	BCEWithLogitsLoss
Optimizer	Adam
Learning rate	0.001
Batch size	32
Number of epochs	200

Table 7 Experimental results of different optimizers.

Optimizer	ACC	AUC	PRE	SP	SN	F1	MCC	Loss	Time
adam	0.9909	0.9985	0.9917	0.9896	0.9921	0.9906	0.9818	0.0417	30.1001
sgd	0.9777	0.9988	0.9713	0.9834	0.9723	0.9773	0.9555	0.1117	24.0389
rmsprop	0.9686	0.9983	0.9956	0.9397	0.9960	0.9668	0.9385	0.0176	27.1115
adamw	0.9605	0.9972	0.9846	0.9335	0.9862	0.9584	0.9220	0.0238	30.7972

loss than SGD. Therefore, we selected Adam as the optimizer for our proposed AttCNN model. Furthermore, a graphical representation of optimizer performance considering both loss and execution time is depicted in Fig. 6.

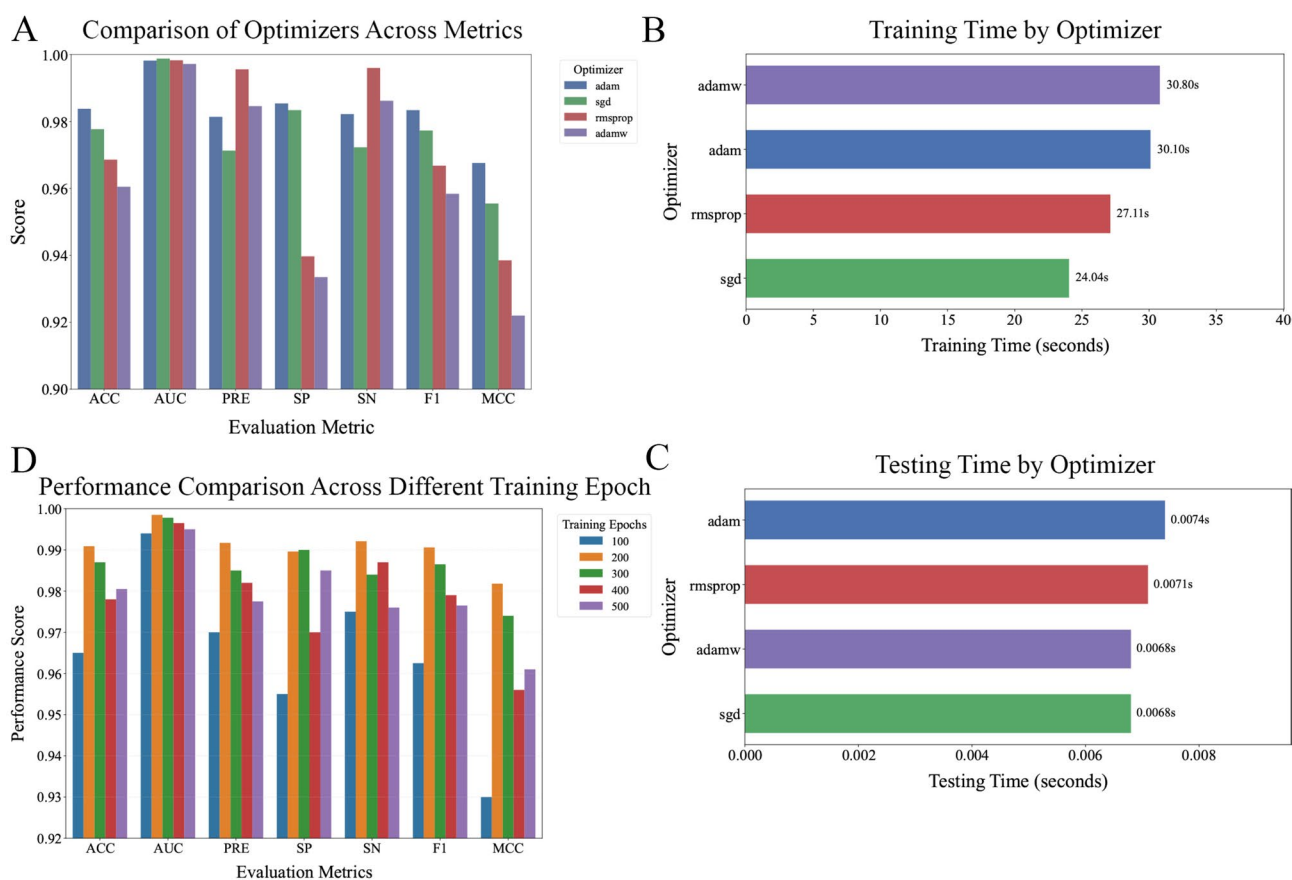
Table 8 illustrates the relationship between learning rate and model performance. When using a learning rate of 0.1, the model failed to train effectively, as the performance metrics did not reach optimal values. An excessively high learning rate may cause convergence instability, which can lead to fluctuations or an increase in the loss. Conversely, a low learning rate may cause the model to get stuck in a local minimum and hinder performance. As presented in Table 8, a learning rate of 0.001 enabled the AttCNN model to achieve superior accuracy and minimized loss. Also, Table 9 shows how the model performed with different epoch settings (100, 200, 300, 400, and 500), indicating that 200 epochs had the biggest positive impact on the model’s ability to make predictions. To facilitate a comprehensive comparison, Fig. 8 presents the performance of the proposed model across various evaluation metrics and training epochs, along with an analysis of its training and testing times.

Table 8 Performance comparison of our proposed model across different learning rates.

Learning rate	ACC	AUC	PRE	SP	SN	F1	MCC
0.1	0.9200	0.9500	0.9300	0.9100	0.9250	0.9180	0.8400
0.01	0.9750	0.9950	0.9800	0.9650	0.9850	0.9740	0.9500
0.001	0.9909	0.9985	0.9917	0.9896	0.9921	0.9906	0.9818
0.0001	0.9600	0.9900	0.9650	0.9500	0.9700	0.9590	0.9200
0.00001	0.9000	0.9300	0.9100	0.8900	0.9150	0.8950	0.8000

Table 9 Evaluation of our proposed model's performance across different training epochs.

Epochs	ACC	AUC	PRE	SP	SN	F1	MCC
100	0.9650	0.9940	0.9700	0.9550	0.9750	0.9625	0.9300
200	0.9909	0.9985	0.9917	0.9896	0.9921	0.9906	0.9818
300	0.9870	0.9978	0.9850	0.9900	0.9840	0.9865	0.9740
400	0.9780	0.9965	0.9820	0.9700	0.9870	0.9790	0.9560
500	0.9805	0.9950	0.9775	0.9850	0.9760	0.9765	0.9610


Fig. 8 Model performance comparison across metrics and training epochs (A,D) and optimizer training and testing time analysis (B,C).

Statistical analysis

In this section, we performed a statistical evaluation utilizing the t-test and the Wilcoxon signed-rank test to examine the significance of our proposed methodology on model performance, as illustrated in Table 10. These statistical analyses were conducted by importing the SciPy stats module⁵⁰ implemented in Python. The resulting p-values for each model and evaluation metric were substantially below the conventional significance level of 0.05, thereby providing strong statistical evidence to reject the null hypothesis and suggesting that the observed performance differences are not attributable to random variation. Specifically, the p-values obtained for the DT, RF, XGBoost, AdaBoost, and AttCNN models were 0.0210, 0.0031, 0.0008, 0.0065, and 0.0003, respectively. Each of these p-values falls below the conventional threshold of 0.05, thereby confirming statistical significance. Notably, the AttCNN model produced a p-value of 0.0003 that is considerably lower than the threshold and indicates a highly statistically significant performance improvement.

Moreover, the proposed AttCNN approach exhibited statistically significant performance improvements over four baseline classifiers: DT, RF, XGBoost, and AdaBoost, across multiple evaluation metrics. Compared to the AdaBoost method, our model demonstrated statistically superior outcomes, as all p-values remained under the 0.05 significance level for each metric considered. In summary, the conducted statistical analyses confirm that our proposed AttCNN model delivers meaningful and consistent performance improvements. These results not only reinforce the validity of our method but also highlight its superiority compared to established state-of-the-art classification approaches.

Web application for ovarian cancer classification

In order to improve the practical utility and accessibility of our proposed AttCNN model, we created a web-based application that is capable of real-time classification of OC from H-imgs. The deployment framework integrates three core elements that collaborate to enable robust inference functionality. At the outset, all trained models, namely the AttResNet50 feature extractor, SHAP feature selector, and AttCNN classifier, are published on the Hugging Face Model Hub, guaranteeing version management, straightforward accessibility and experimental reproducibility⁵¹. Using these models as a foundation, we engineered a RESTful API with the FastAPI framework in Python 3.10, subsequently deployed on Hugging Face Spaces through Docker SDK^{52,53}. This API manages the complete computational workflow, including model initialization, image preprocessing operations, feature extraction using AttResNet50, SHAP-based feature selection validated in our experimental procedures, and final classification executed by the AttCNN. To maximize user accessibility and ease of use, we created a client-side interface utilizing standard web technologies served by FastAPI, with Vercel hosting the deployment. This interface provides users with the capability to upload H-imgs through a streamlined browser environment, establishing communication with the Hugging Face Space API to obtain predictions. All source code from the experimental work is accessible via the GitHub repository at <https://github.com/camlas/ovarian-cancer-histopathology>.

Table 10 Statistical analysis of feature extractors and classifiers using evaluation metrics (metric-wise) and classifier performance (model-wise).

Metric-wise	Metric				Model-wise	Model			
Metric	Min	Max	Mean	Std	Model	Min	Max	Mean	Std
ACC	0.6798	0.9878	0.8953	0.1102	DT	0.6798	0.8176	0.7690	0.0663
AUC	0.6766	0.9995	0.9390	0.0975	RF	0.9108	0.9956	0.9561	0.0325
PRE	0.7273	0.9916	0.9036	0.1021	XGBoost	0.9319	0.9664	0.9495	0.0147
SP	0.5489	0.9938	0.8780	0.1383	AdaBoost	0.9360	0.9938	0.9611	0.0249
SN	0.8043	0.9921	0.9373	0.0756	AttCNN	0.9625	0.9876	0.9793	0.0112
F1	0.6256	0.9876	0.9156	0.1069					
MCC	0.3661	0.9757	0.8585	0.1632					

Table 11 Comparison of our proposed model with existing state-of-the-art models.

Article author	ACC	AUC	PRE	SP	SN	F1
Mahmoudiandehkordi, Shaghayegh et al. ⁵⁴	0.9811	–	0.9914	–	0.9888	0.9879
Gao Y, Zeng S et al. ²⁵	0.772	–	–	0.749	0.875	0.583
Zhang K, Qiu Y et al. ⁵⁵	–	0.889	0.94	–	0.829	0.823
Singh S, Maurya MK et al. ³³	0.9810	0.97	0.96	0.96	0.96	–
Proposed model	0.9909	0.9985	0.9917	0.9896	0.9921	0.9906

Comparison with existing methods

In this work, we thoroughly compared our suggested AttCNN model with a number of current cutting-edge methods created for OC classification on the used dataset. To rigorously evaluate the performance and generalizability of our model, multiple evaluation criteria were employed, including ACC, AUC, precision, specificity, sensitivity, and the F1 score. The benchmarking was carried out against methods introduced by Mahmoudiandehkordi, Shaghayegh et al.⁵⁴, Gao Y, Zeng S et al.²⁵, Zhang K, Qiu Y et al.⁵⁵, and Singh S, Maurya MK et al.³³, as detailed in Table 11.

In addition, our model achieved precision, recall, and F1 scores of 0.9917, 0.9921, and 0.9906, respectively, all of which exceed the performance reported by existing models. These results clearly demonstrate the effectiveness and reliability of the proposed framework for detecting OC. To optimize performance, we conducted extensive experiments with various CNN architectures and ML classifiers to identify the most effective combination of feature extractor and classification method. A comprehensive analysis of optimization algorithms was also performed for the AttCNN model, enabling the selection of an optimizer that improved convergence speed and minimized training loss. Furthermore, we evaluated multiple feature selection techniques and identified SHAP as the most effective based on empirical results. The integration of the attention mechanism with CNNs and the selected features further enhanced the model's ability to capture meaningful patterns in the data. This hybrid design leverages the strengths of each individual component, resulting in a model that consistently outperforms existing cutting-edge methods on this dataset. Consequently, our proposed AttCNN framework establishes a new benchmark for robust and accurate OC classification.

Conclusion

The diagnosis of OC using H-imgs is a complex and time-intensive process, largely due to the limited availability of skilled pathologists and the manual effort required throughout the procedure. In this study, we proposed a resilient framework that integrates ML and DL techniques with an attention mechanism to identify OC from histopathological slides. After evaluating various denoising techniques, we chose the bilateral filtering approach for transforming the input image, as it demonstrated superior classification performance compared to the other methods. We implemented data augmentation and normalization during the preprocessing stage to facilitate learning from diverse representations and to speed up convergence. A modified version of ResNet50 was utilized for feature extraction, where transfer learning was applied and the model was fine-tuned through parameter optimization. We utilized SHAP, a feature selection technique based on cooperative game theory, due to its effectiveness in precisely identifying important features within the test dataset. Furthermore, we selected AttCNN for classification among five widely recognized ML algorithms due to its superior performance on our dataset. The experimental results show that our model performs better at predicting outcomes than the current state-of-the-art methods for classifying OC cancer. We further built a web-based application that integrates our proposed model for the early detection of OC in clinical environments. Despite these encouraging findings, the present study has several limitations. The proposed framework was validated using only a single OC dataset, which may restrict its generalizability across diverse clinical settings and histopathological imaging conditions.

In future work, we intend to evaluate the framework on multiple large-scale and multi-center datasets to further examine its robustness and clinical applicability. Additionally, Vision Transformer (ViT)-based architectures will be explored to improve feature extraction and representation learning capabilities. We also plan to incorporate uncertainty-aware analysis and outlier detection strategies, including pLOF-based techniques, to improve the reliability and interpretability of the predictive framework.

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Data availability The datasets used in the current study are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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